Lesson notes · for you

Blanks, repeats and impossible values

Data engineering · Level 2 · Lesson 3

Time	Materials
15 min	Your phone with Darsi, for the Python steps; A table you made, or a made-up one, and a pen

Learning goals

- Tell a blank from a zero; never guess.
- Find repeats and impossible values.
- Write a Python check for blanks and values out of range.

Activity: Spot the problems

1. Look (3 min): Look at a made-up rain table with a 999, a blank, a minus sign and a repeat hidden in it.
2. Hunt (5 min): In pairs, circle every problem you can find.
3. Decide (4 min): Next to each problem, write what to do: say it's missing, keep one copy, or ask.
4. Compare (3 min): Work out the average with and without the problems. How much did it change?

Keep the numbers small so the averages are quick to work out by hand.

Discussion

- Why is a blank not the same as a zero?
- When is a repeat not really a repeat?
- Why check for blanks before checking the range?

Check yourself

- Write one blank, one repeat and one impossible value, and what you'd do with each.

Try it this week

Check a table you made, or one from class. Circle every blank, every repeat and every impossible value. For each one, write what you'd do.

Worksheet

Blanks, repeats and impossible values

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Name _____ Class _____ Date _____

1 **A mark says 25, but the test was out of 20. What do you do?**

◯ Keep it: the pupil did very well ◯ Change it to 20 ◯ Check it with the teacher: it's impossible

2 **Zero rain and a blank rain reading mean the same thing.**

◯ True ◯ False

3 **None means a blank mark. Fix the program so it counts the blanks, not the zeros: it should print 2.**

```
marks = [14, None, 17, 0, None, 12]
blanks = 0
for m in marks:
    if m == 0:
        blanks = blanks + 1
print(blanks)
```

Write your program:

4 **This should flag 25 and -2, but it prints nothing. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  marks = [14, 25, 9, -2, 18]
2  for m in marks:
3      if m < 0 and m > 20:
4          print(m, "check")
```

5 **Now both checks together. Run it: it crashes on a blank. Fix it so it skips the blanks and counts them, prints each impossible mark with "check", then prints how many blanks there were.**

```
marks = [14, None, 25, 9, -2, None, 18]
blanks = 0
for m in marks:
    if m < 0 or m > 20:
        print(m, "check")
print(blanks, "blank")
```

Write your program:

Have you ever seen a number that couldn't be right, in a game score, a report or a chart? What gave it away?

Check your answers

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1 **Check it with the teacher: it's impossible**

A mark can't be above the maximum, so someone made a mistake.

2 **False**

Zero means someone read the gauge and it was dry.

3

```
marks = [14, None, 17, 0, None, 12]
blanks = 0
for m in marks:
    if m is None:
        blanks = blanks + 1
print(blanks)
```

The starter counted the one real zero. m is None is true only for the two blanks, so it prints 2.

4 **Line 3: `if m < 0 or m > 20:`**

No number is below 0 and above 20 at the same time, so the and test is never true.

5

```
marks = [14, None, 25, 9, -2, None, 18]
blanks = 0
for m in marks:
    if m is None:
        blanks = blanks + 1
    elif m < 0 or m > 20:
        print(m, "check")
print(blanks, "blank")
```

Blanks come first, because you can't compare nothing with a number.